

Network Security Laboratory Project

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Executive Summary

This project presents practical network security labs covering traffic analysis, network scanning, secure protocols, and firewall configuration.

Introduction

Hands-on exercises were conducted using VirtualBox, Kali Linux, Wireshark, Nmap, OpenSSL, SSH/SFTP, and UFW.

Project Objectives

- Build a virtual network
- Analyze traffic
- Perform enumeration
- Compare HTTP/HTTPS and FTP/SFTP
- Configure firewalls

Laboratory Environment

Tools: VirtualBox, Kali Linux, Wireshark, Nmap, OpenSSL, SSH/SFTP, UFW

Phase 1 – Network Setup and Connectivity Verification

```
(salem@vbox)-[~]
$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:dc:75:ca brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
        valid_lft 83881sec preferred_lft 73081sec
    inet6 fd00::6640:64c6:ed88:c5f2/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 84898sec preferred_lft 12898sec
    inet6 fe80::29ad:522a:3ce1:92b/64 scope link
        valid_lft forever preferred_lft forever
3: eth1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:57:a2:01 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.108/24 brd 192.168.56.255 scope global dynamic noprefixroute eth1
        valid_lft 477sec preferred_lft 477sec
    inet6 fe80::a00:27ff:fe57:a201/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

(salem@vbox)-[~]
$ ping -c 4 192.168.56.107
PING 192.168.56.107 (192.168.56.107) 56(84) bytes of data.
64 bytes from 192.168.56.107: icmp_seq=1 ttl=64 time=0.388 ms
64 bytes from 192.168.56.107: icmp_seq=2 ttl=64 time=0.598 ms
64 bytes from 192.168.56.107: icmp_seq=3 ttl=64 time=0.753 ms
64 bytes from 192.168.56.107: icmp_seq=4 ttl=64 time=0.718 ms

— 192.168.56.107 ping statistics —
4 packets transmitted, 4 received, 0% packet loss, time 3029ms
rtt min/avg/max/mdev = 0.388/0.614/0.753/0.142 ms
```

```

ziad@ziad-VirtualBox:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group defa
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 08:00:27:c0:60:8d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
        valid_lft 82085sec preferred_lft 82085sec
    inet6 fd00::7ede:ce2f:2795:6400/64 scope global temporary dynamic
        valid_lft 85897sec preferred_lft 13897sec
    inet6 fd00::fceb:3c79:b371:4219/64 scope global dynamic mngtmpaddr noprefi
        valid_lft 85897sec preferred_lft 13897sec
    inet6 fe80::7a05:8103:a60a:3078/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP
    link/ether 08:00:27:af:73:ce brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.107/24 brd 192.168.56.255 scope global dynamic noprefixrou
        valid_lft 485sec preferred_lft 485sec
    inet6 fe80::c885:3a74:9eea:f80d/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
4: virbr0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DO
    link/ether 52:54:00:9e:a7:1e brd ff:ff:ff:ff:ff:ff
    inet 192.168.122.1/24 brd 192.168.122.255 scope global virbr0
        valid_lft forever preferred_lft forever
ziad@ziad-VirtualBox:~$ ping -c 4 192.168.56.108
PING 192.168.56.108 (192.168.56.108) 56(84) bytes of data.
64 bytes from 192.168.56.108: icmp_seq=1 ttl=64 time=0.814 ms
64 bytes from 192.168.56.108: icmp_seq=2 ttl=64 time=1.73 ms
64 bytes from 192.168.56.108: icmp_seq=3 ttl=64 time=1.71 ms
64 bytes from 192.168.56.108: icmp_seq=4 ttl=64 time=1.55 ms

--- 192.168.56.108 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3078ms
rtt min/avg/max/mdev = 0.814/1.450/1.727/0.373 ms

```

No.	Time	Source	Destination	Protocol	Length	Info
9	47.723639458	192.168.56.108	192.168.56.107	ICMP	98	Echo (ping) request id=0x:
10	47.724046870	192.168.56.107	192.168.56.108	ICMP	98	Echo (ping) reply id=0x:
11	48.796688666	192.168.56.108	192.168.56.107	ICMP	98	Echo (ping) request id=0x:
12	48.797356536	192.168.56.107	192.168.56.108	ICMP	98	Echo (ping) reply id=0x:
13	49.808248492	192.168.56.108	192.168.56.107	ICMP	98	Echo (ping) request id=0x:
14	49.809039061	192.168.56.107	192.168.56.108	ICMP	98	Echo (ping) reply id=0x:
15	50.810415923	192.168.56.108	192.168.56.107	ICMP	98	Echo (ping) request id=0x:
16	50.811040437	192.168.56.107	192.168.56.108	ICMP	98	Echo (ping) reply id=0x:

Wireshark ICMP Capture

Phase 2 – Network Scanning and Enumeration

```

salem@vbox: ~
Session Actions Edit View Help
(salem@vbox)-[~]
$ nmap -sn 192.168.56.0/24
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-09 21:19 -0400
Nmap scan report for 192.168.56.1 (192.168.56.1)
Host is up (0.00078s latency).
MAC Address: 0A:00:27:00:00:10 (Unknown)
Nmap scan report for 192.168.56.100 (192.168.56.100)
Host is up (0.0023s latency).
MAC Address: 08:00:27:21:00:55 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.107 (192.168.56.107)
Host is up (0.00082s latency).
MAC Address: 08:00:27:AF:73:CE (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.56.108 (192.168.56.108)
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 2.05 seconds

(salem@vbox)-[~]
$ sudo nmap -sS -p 1-1000 192.168.56.107
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-09 21:19 -0400
Nmap scan report for 192.168.56.107 (192.168.56.107)
Host is up (0.0019s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
MAC Address: 08:00:27:AF:73:CE (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 0.41 seconds

(salem@vbox)-[~]
$ sudo nmap -sV 192.168.56.107
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-09 21:19 -0400
Nmap scan report for 192.168.56.107 (192.168.56.107)
Host is up (0.0014s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)
MAC Address: 08:00:27:AF:73:CE (Oracle VirtualBox virtual NIC)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 1.11 seconds

(salem@vbox)-[~]
$
```

nmap host discovery syn scan

```

(salem@vbox)-[~]
$ sudo nmap -O 192.168.56.107
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-09 21:20 -0400
Nmap scan report for 192.168.56.107 (192.168.56.107)
Host is up (0.0012s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
MAC Address: 08:00:27:AF:73:CE (Oracle VirtualBox virtual NIC)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 2.05 seconds

(salem@vbox)-[~]
$ sudo nmap -A 192.168.56.107 -oN nmap_week2_report.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-09 21:21 -0400
Nmap scan report for 192.168.56.107 (192.168.56.107)
Host is up (0.0011s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 04:ec:33:95:8a:f9:2c:ac:be:f9:1a:af:7d:ca:44:f3 (ECDSA)
|_  256 3b:f1:ca:6e:53:11:3f:94:e1:33:56:a7:7d:25:07:3f (ED25519)
MAC Address: 08:00:27:AF:73:CE (Oracle VirtualBox virtual NIC)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   1.12 ms  192.168.56.107 (192.168.56.107)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 6.26 seconds

```

nmap host discovery os fingerprint report

```

(salem@vbox)-[~]
$ cat nmap_week2_report.txt
# Nmap 7.99 scan initiated Sat May  9 21:21:04 2026 as: /usr/lib/nmap/nmap -A -oN nmap_week2_report.txt 192.168.56.107
Nmap scan report for 192.168.56.107 (192.168.56.107)
Host is up (0.0011s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 04:ec:33:95:8a:f9:2c:ac:be:f9:1a:af:7d:ca:44:f3 (ECDSA)
|_  256 3b:f1:ca:6e:53:11:3f:94:e1:33:56:a7:7d:25:07:3f (ED25519)
MAC Address: 08:00:27:AF:73:CE (Oracle VirtualBox virtual NIC)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

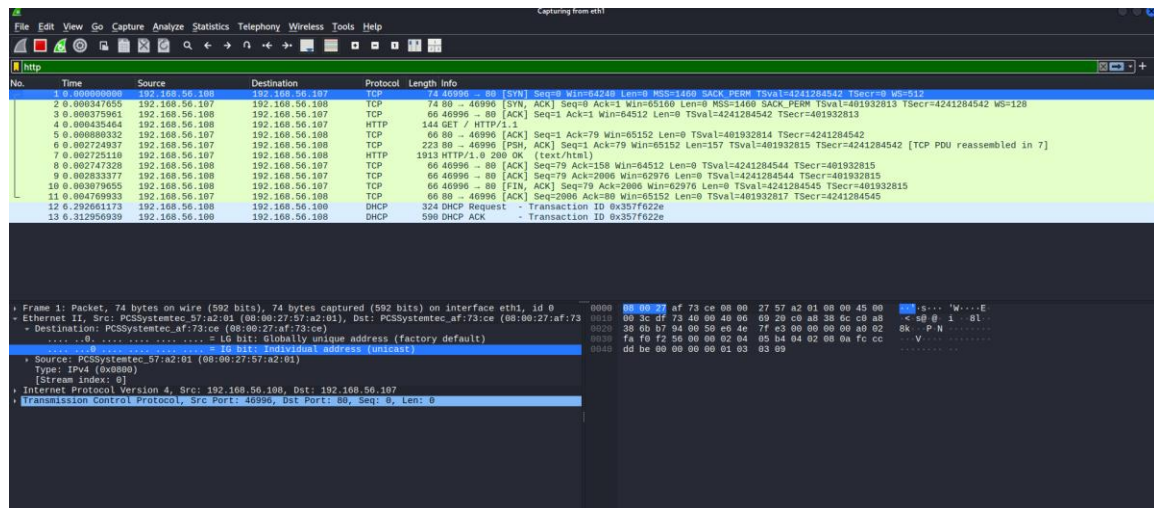
TRACEROUTE
HOP RTT      ADDRESS
1   1.12 ms  192.168.56.107 (192.168.56.107)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
# Nmap done at Sat May  9 21:21:10 2026 -- 1 IP address (1 host up) scanned in 6.26 seconds

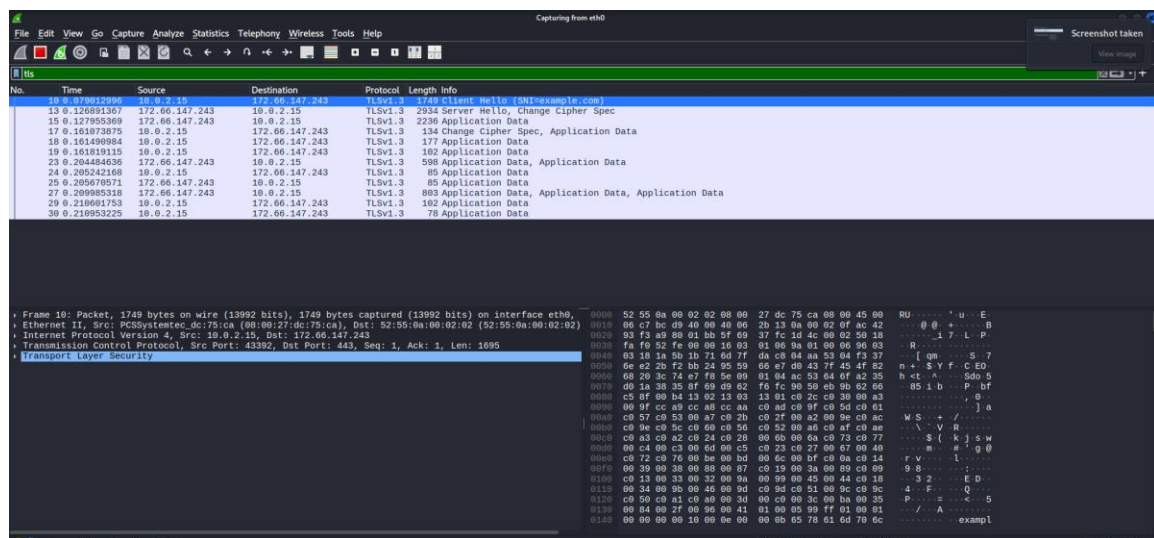
```

Nmap report file

Phase 3 – Packet Capture and Traffic Analysis



wire shark HTTP Traffic



wireshark HTTPS Traffic

Phase 4 – Secure Communication Protocols

```
CONNECTED(00000003)
depth=3 C=US, O=SSL Corporation, CN=SSL.com TLS ECC Root CA 2022
verify return:1
depth=2 C=US, O=SSL Corporation, CN=SSL.com TLS Transit ECC CA R2
verify return:1
depth=1 C=US, O=CLOUDFLARE, INC., CN=Cloudflare TLS Issuing ECC CA 1
verify return:1
depth=0 CN=example.com
verify return:1
—
Certificate chain
 0 s:CN=example.com
  i:C=US, O=CLOUDFLARE, INC., CN=Cloudflare TLS Issuing ECC CA 1
  a:PKEY: EC, (prime256v1); sigalg: ecdsa-with-SHA256
  v:NotBefore: Apr  2 21:18:57 2026 GMT; NotAfter: Jul  1 21:24:46 2026 GMT
 1 s:C=US, O=CLOUDFLARE, INC., CN=Cloudflare TLS Issuing ECC CA 1
  i:C=US, O=SSL Corporation, CN=SSL.com TLS Transit ECC CA R2
  a:PKEY: EC, (prime256v1); sigalg: ecdsa-with-SHA384
  v:NotBefore: Oct 31 17:17:49 2023 GMT; NotAfter: Oct 28 17:17:48 2033 GMT
 2 s:C=US, O=SSL Corporation, CN=SSL.com TLS Transit ECC CA R2
  i:C=US, O=SSL Corporation, CN=SSL.com TLS ECC Root CA 2022
  a:PKEY: EC, (secp384r1); sigalg: ecdsa-with-SHA384
  v:NotBefore: Oct 21 17:02:23 2022 GMT; NotAfter: Oct 17 17:02:22 2037 GMT
 3 s:C=US, O=SSL Corporation, CN=SSL.com TLS ECC Root CA 2022
  i:C=GB, ST=Greater Manchester, L=Salford, O=Comodo CA Limited, CN=AAA Certificate Services
  a:PKEY: EC, (secp384r1); sigalg: sha256WithRSAEncryption
  v:NotBefore: Aug  1 00:00:00 2025 GMT; NotAfter: Dec 31 23:59:59 2028 GMT
—
Server certificate
——BEGIN CERTIFICATE——
MIID0jCCA3mgAwIBAgIQZSBYnvF+tVxmRDPyny5oSjAKBggqhkJOPQQDAjBSMQsw
CQYDVQQGEwJVUzEZMBcGA1UECgwQQ0xPVURGTFRSR5SwSU5DLjEoMCYGA1UEAwf
Q2xvdWRmbGFyZSBUTFMgSKNzdWluZyBFQ0MgQ0EgMTAeFw0yNjA0MDIyMTE4NTda
Fw0yNjA3MDEyMTI0NDZaMBYxZDASBgNVBAMMC2V4YW1wbGUuY29tMFkwEwYHKoZI
zj0CAQYIKoZIzj0DAQcDQgAE6Iftpk9H1u+tKg0vUFy8V3hSE4MNpwN2wXYtLt8
cUe8Ih2Nvzv6ejX+8cFwk+1SezvPlnoSR/PZmiA26sxVXKOCAMswggJnMAwGA1Ud
EwEB/wQCMAAwHwYDVR0jBBBgFoAUnMQJckcYF3unGomzkjXV4QOM/pIwbAYIKwYB
BQUHAQEYEBMDkGCCsGAQUFBzAChilodHRwOi8vaS5jZi1iLnNzbC5jb20vQ2xv
dWRmbGFyZS1UTFMtSS1FMS5jZlIwIQYIKwYBBQUHMAAGFWh0dHA6Ly9vLmNmLWlU
c3NsLmNvbTA1BgNVHREEHjAcggtleGFtcGxlLmNvbYINKi5leGFtcGxlLmNvbTAj
BgNVHSAEHDAaMAgGBmeBDAECATA0BgwrBgEEAYKpMAEDAQEWYDVR0LBAwwCgYI
KwYBBQUHAWEPgYDVR0fBDcwNTAzoDGgL4YtaHR0cDovL2MuY2YtYi5zc2wuY29t
L0Ns3VkbZmxhcmUtVEXtLUktRTEuY3JsMA4GA1UdDwEB/wQEAWIHgDAPBgkrBgEE
AYLaSywEAGUAMIIBBAYKKwYBBAHWeQIEAgSB9QSB8gDwAHYAyKPEf8ezrbk1awE/
anoSbeM6Tk0Lxkb5l605dZkdz5oAAAGdUBlyrwAABAMARzBFAiB/1tZdXGNf67TC
SzLFxka7y1mz2gEpXm03uk2SVPNANQIhAPw/Quf0Ye0h7aK6BaalgoisAB8t/Fbp
QP/W7EMFRDhPAHYALE5Dh/rswe+B8xkkJqgYZQH0184AgE/cmd9VTcuGdgAAAGd
UBlykAAABAMARzBFAiABZgcvmVAZLISG7fvc5NYPdvJD97noT6GthzaZI1phEQIh
AJFKuSjsdurepc2e2QBrSdghAGQzMYVaSomJ++/1bbnIMAoGCCqGSM49BAMCA0cA
MEQCIHY8XBGm9RDpvUkDnmTADfe/IJrDV/vwKB5oNEKb2x4AiB8OS8DiJ2H6pc/
zKbrdX+gLU4C15Dsggh/2v2xSoPdIA=
——END CERTIFICATE——
```

openssl tls certificate

```
-----END CERTIFICATE-----
subject=CN=example.com
issuer=C=US, O=CLOUDFLARE, INC., CN=Cloudflare TLS Issuing ECC CA 1
-----
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: ecdsa_secp256r1_sha256
Negotiated TLS1.3 group: X25519MLKEM768
-----
SSL handshake has read 5052 bytes and written 1758 bytes
Verification: OK
-----
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Protocol: TLSv1.3
Server public key is 256 bit
This TLS version forbids renegotiation.
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 0 (ok)
-----
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol : TLSv1.3
    Cipher : TLS_AES_256_GCM_SHA384
    Session-ID: F9D319F92DD364C1F344C8D0A40B668AE79833BF88A7B8933085EA1A831915F2
    Session-ID-ctx:
    Resumption PSK: 1611A5837B821BD61E4220BF51A38571B61DB94DD647C1AA1D9E5C5EE2C81E4E3143A0926AA1721F380103470D95B886
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 64800 (seconds)
    TLS session ticket:
0000 - 8c a1 5d 41 d1 22 0a 9e-2d 9a fa d3 a9 96 c2 f9 ..]A."..-.....
0010 - 60 a6 fc 69 f8 4e 70 58-61 a1 ac 4a 98 45 63 69 `..i.NpXa..J.Eci
0020 - e1 30 47 3e 4e ad 0d 50-1e b0 73 73 3a c8 b1 be .0G>N..P..ss: ...
0030 - 3b d8 c7 e4 9e 63 c5 74-f3 8c aa e3 5c e2 f1 0c ;....C.t....\...
0040 - 9f 5c 0e ec 98 36 04 1b-c8 d9 d3 94 8f b2 92 36 .\...6.....6
0050 - a2 f3 4b 4e 86 2b a7 56-b6 e8 ba 76 91 cb c0 f7 ..KN.+..V...v....
0060 - 56 63 c0 50 d7 c9 4a b6-84 d9 ca 5b e5 02 7b 52 Vc..P..J....[..{R
0070 - 3e 4c 7a 40 d1 97 9b a9-37 f9 8b f4 48 a6 4c c0 >Lz@....7...H.L.
0080 - 7d 87 a7 c8 d2 be 04 a0-72 69 c3 95 7b 7f 22 15 }......ri..{.".
0090 - 8b 34 43 a2 5d d0 08 4c-08 ed c1 e6 32 66 06 10 .4C.]..L....2f..
00a0 - e8 06 df e2 2a 0a 64 39-60 83 cb 9a 62 fc 7d ed ....*.d9`...b.}.
00b0 - ea 64 85 6a 57 f1 eb c4-9e 67 5b ca ab d2 aa 87 .d.jW....g[.....

    Start Time: 1778418177
    Timeout : 7200 (sec)
    Verify return code: 0 (ok)
    Extended master secret: no
```

openssl tls session

```

Start Time: 1778418177
Timeout : 7200 (sec)
Verify return code: 0 (ok)
Extended master secret: no
Max Early Data: 0

---
read R BLOCK
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol : TLSv1.3
    Cipher : TLS_AES_256_GCM_SHA384
    Session-ID: 05786334D542BCB56802D89FBBFA70A4EACC925892D98D13FE1BDF04AD2902E
    Session-ID-ctx:
    Resumption PSK: 930C075A7BF779273718179855598ECA1C1877D675B92154E7FFDB7EE2F5682B36812A5BE17D133AD8AB2DF9E70C9D9C
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 64800 (seconds)
    TLS session ticket:
0000 - 8c a1 5d 41 d1 22 0a 9e-2d 9a fa d3 a9 96 c2 f9 ..JA."..-.....
0010 - c0 17 1b 6d 01 91 95 de-aa b0 d2 9c 7b 42 f8 dc ...m.....{B..
0020 - f3 67 da f5 0d ae 00 1b-3a 8e 9e fc c0 ea 05 7b .g.....:.....{
0030 - 47 bd 8d 7f 5d 00 4b 63-9f 26 cd 27 96 65 0c 83 G...].Kc.6.'.e..
0040 - 92 ae 3e f4 1a 95 05 9a-0b a8 df 84 7b 08 18 3f ..>.....{..?
0050 - 7e 62 c5 76 c4 af 5e ff-5f 30 6d 4d 7c cb 63 ce ~b.v..^._0mM|.c.
0060 - 4e 23 48 cf c9 ef c2 f8-8e 96 e8 d6 80 ec ff a4 N#H.....
0070 - a7 1d 1c 8e 04 ce e7 a2-6c e0 79 2e af 50 06 1e .....l.y..P..
0080 - 93 70 ab 67 c4 97 18 55-fa 11 cd ef 1c 99 50 a8 .p.g...U.....P.
0090 - b7 b7 4f b3 47 1c 0e 5f-35 b0 52 cc c0 3a ed 45 ..0.G..._5.R...:E
00a0 - 98 7e c0 e4 2d b2 35 3c-1a 03 7c 6a 58 67 ba cf ~..-5<..|jXg..
00b0 - 80 9f 9f ff 03 d3 10 d1-86 bb 21 b6 b1 3a 29 2f .....!..:)/

Start Time: 1778418177
Timeout : 7200 (sec)
Verify return code: 0 (ok)
Extended master secret: no
Max Early Data: 0

---
read R BLOCK
closed

```

openssl tls session2

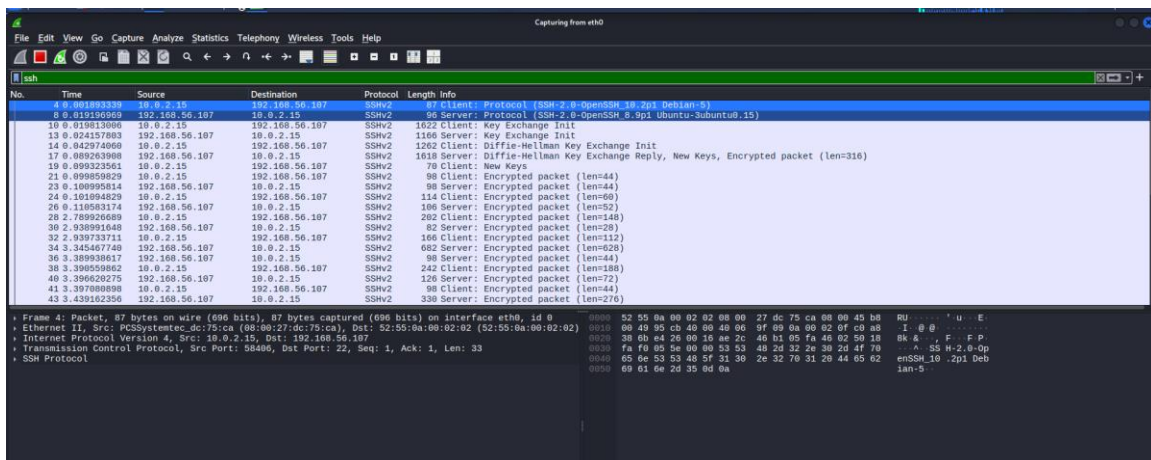
No.	Time	Source	Destination	Protocol	Length	Info
5	0.008103486	192.168.56.107	10.0.2.15	FTP	74	Response: 220 (vsFTPD 3.0.5)
9	3.255624920	10.0.2.15	192.168.56.107	FTP	65	Request: USER ziad
11	3.256576710	192.168.56.107	10.0.2.15	FTP	88	Response: 331 Please specify the password.
13	6.713054156	10.0.2.15	192.168.56.107	FTP	69	Request: PASS ziad1234
15	6.814141310	192.168.56.107	10.0.2.15	FTP	77	Response: 230 Login successful.
17	6.814520125	10.0.2.15	192.168.56.107	FTP	60	Request: SYST
19	6.815851400	192.168.56.107	10.0.2.15	FTP	73	Response: 215 UNIX Type: L8
20	6.816012957	10.0.2.15	192.168.56.107	FTP	60	Request: FEAT
21	6.817076114	192.168.56.107	10.0.2.15	FTP	90	Response: 211-Features:
22	6.817464442	192.168.56.107	10.0.2.15	FTP	98	Response: PASV
24	12.720293494	10.0.2.15	192.168.56.107	FTP	62	Request: TYPE I
26	12.721132145	192.168.56.107	10.0.2.15	FTP	85	Response: 200 Switching to Binary mode.
27	12.721250126	10.0.2.15	192.168.56.107	FTP	60	Request: EPSV
28	12.722181920	192.168.56.107	10.0.2.15	FTP	102	Response: 229 Entering Extended Passive Mode (59048)
32	12.723656507	10.0.2.15	192.168.56.107	FTP	71	Request: STOR sample.txt

wireshark ftp plaintext credentials


```
(salem@vbox)-[~]
$ ftp 192.168.56.107
Connected to 192.168.56.107.
220 (vsFTPD 3.0.5)
Name (192.168.56.107:salem): ziad
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> put sample.txt
local: sample.txt remote: sample.txt
229 Entering Extended Passive Mode (|||59048|)
550 Permission denied.
ftp> bye
221 Goodbye.

(salem@vbox)-[~]
$ sftp ziad@192.168.56.107
The authenticity of host '192.168.56.107 (192.168.56.107)' can't be established.
ED25519 key fingerprint is: SHA256:NGIqBBig2fg9u/sLxClhVHBaoEwbAlgPDyC3w0bMUfs
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.56.107' (ED25519) to the list of known hosts.
ziad@192.168.56.107's password:
Connected to 192.168.56.107.
sftp> put sample.txt
Uploading sample.txt to /home/ziad/sample.txt
sample.txt
sftp> █
```

ftp vs sftp



wireshark sftp encrypted

Phase 5 – Firewall Configuration

```
ziad@ziad-VirtualBox:~$ sudo ufw enable
sudo ufw default deny incoming
sudo ufw allow 22/tcp
sudo ufw deny 80/tcp
sudo ufw status verbose
Firewall is active and enabled on system startup
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
Rule added
Rule added (v6)
Rule added
Rule added (v6)
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), deny (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
80/tcp DENY IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) DENY IN Anywhere (v6)

ziad@ziad-VirtualBox:~$
```

UFW Rules

```
Session Actions Edit View Help
(salem@vbox)-[~]
$ curl -I http://192.168.56.107
curl: (7) Failed to connect to 192.168.56.107 port 80 after 21005 ms: Could not connect to server

(salem@vbox)-[~]
$ ssh ziad@192.168.56.107
ziad@192.168.56.107's password:
Permission denied, please try again.
ziad@192.168.56.107's password:
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-111-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

378 updates can be applied immediately.
296 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

ziad@ziad-VirtualBox:~$
```

Firewall Validation

Key Findings

HTTPS protects traffic, SFTP is more secure than FTP, and firewall rules effectively control access.

Skills Demonstrated

Wireshark, Nmap, TLS Analysis, Linux Security, Firewall Administration, Network Security.

Conclusion

The project demonstrates core network security concepts and practical defensive skills.